

Basics of git for the Project

- **How GitHub works**

GitHub is a website used to share code among people, providing structure and tools to avoid any issue. A repository (where the code is) can be split into different branches. Each user works on a local version of this repository before committing their code with the others (that is sharing their work on the code). Each branch of the repository represents a different version of the project (different files or same file but different improvements or timeline). When a user commits their code to the repository they commit it towards the branch they are on. More on branches if need be. The original branch is called main, it is where the final project is supposed to be. The advantage of github is that it provides a way to make sure that we can share the code on branches safely (without erasing each other's code). The way this works is that before a user can work on code, they must pull from main what has been done (that is get locally the latest version of the shared repository) to then work and finally push their version (upload their code to the shared repository). We will go through the commands to do that later.

- **How to use it**

I would recommend using github on your terminal (on Mac click command and space bar then spell terminal). It is also good to know that you can have a terminal on VScode (which btw if you don't know what app to use to code i would also recommend). We will see if we need branches in the future (pretty sure we will). If we do i'll fill you guys in on how they work and what they really imply). Git works using commands in your terminal in your local repository

- **How to clone the repository locally**

Your local repository (clone) is where you will write your code. It is the one you can update to its latest version and where you can upload your code to the shared repository all using git. It is basically a folder that you have on your computer and that contains all the files of the shared repository. To have it on your computer we will use the terminal. go to

wherever you want on your computer (would recommend the desktop if you are not sure) (you can navigate your computer by using the command “`cd folder_name`” to go inside a folder named *folder_name* and the command “`ls`” (thats an L) to see what is inside that folder). Now go on the repository for T4D4 and click on the green button code, click on https and copy the *link* given. Where you want to put your folder use in your terminal the command “`git clone link`”. You should see the local clone repository appear. To go inside that folder again use the command “`cd T4D4`”

- **Useful commands (thats whats important)**

As mentioned we use commands in the terminal to use gits functionalities, here are the ones (in order of use) that you need to know

- ❖ “git fetch” : to see if there is anything new on the shared repository (always remember to check if you have the latest version before working on things that is very important)
- ❖ “git pull” : if it happens that you don’t have the latest version locally then git pull will make sure of that
- ❖ “git add .” : adds to your commit all the new files you created
- ❖ “git commit -m “*message*” ” : to commit your changes along with the commit *message* that explains what explains what you have changed or added
- ❖ “git push” : to send your commits to the shared repository to update it with your work for everyone to enjoy

All of this should cover the basics, I tried to be as exhaustive yet concise as possible. If you have any questions, don’t hesitate and keep in mind that I would rather answer the same question 1000 times than deal with merging issues :)